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Research Paper



Thermodynamic performance and optimization of the helium-xenon Brayton cycle for gas-cooled micro-reactors

Wenkui Ma, Xiaoyong Yang *, Jie Wang

Institute of Nuclear and New Energy Technology of Tsinghua University, Collaborative Innovation Center of Advanced Nuclear Energy Technology, The Key Laboratory of Advanced Reactor Engineering and Safety, Ministry of Education, Beijing 100084 China

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ABSTRACT

The gas-cooled micro-reactor Brayton cycle using helium-xenon working fluid holds significant potential for applications in distributed energy supply, power assurance in remote areas, and other fields. However, the system's thermodynamic performance under engineering constraints has not been sufficiently explored. This study developed thermodynamic models and identified the primary factors influencing cycle performance, including system parameters and working fluid properties. The effects of key parameters on system performance were investigated. The results indicated that an increased helium fraction within the helium-xenon working fluid leads to changes in physical properties, resulting in an increase in turbine output work, cycle specific power generation, and cycle efficiency. Additionally, lower pressure ratios and higher recuperation effectiveness enhance cycle efficiency but may lead to the reactor pressure vessel temperature exceeding the limit 823 K. Therefore, optimal thermal design schemes were obtained through dual-objective optimization, considering the reactor pressure vessel temperature constraint. For helium, 40 $g \bullet mol^{-1}$ helium-xenon, 80 $g \bullet mol^{-1}$ helium-xenon, and xenon working fluids, the cycle efficiencies of the optimal schemes under the reactor pressure vessel temperature constraint are 37.32 %, 36.16 %, 32.63 %, and 24.31 %, respectively. This study provides a reference for the design and optimization of gas-cooled micro-reactor Brayton cycle.

1. Introduction

Amid the growing global energy demand, the search for efficient, safe, and sustainable energy systems has become a key focus of scientific research. Nuclear energy, with its high energy density and low carbon emissions, offers significant advantages in addressing climate change and reducing greenhouse gas emissions. As an emerging nuclear technology, gas-cooled micro-reactors (GCMR) have garnered increasing attention in recent years due to their unique design and broad application prospects. Compared to traditional nuclear reactors, GCMR feature a small size, compact structure, and ease of modular deployment, making them highly promising for applications such as distributed energy supply, power assurance in remote areas, and emergency power support [1,2].

The Brayton cycle employs reactor coolant as the working fluid, fully leveraging the high outlet temperature of the reactor. With advantages such as high cycle efficiency and chemically inert coolant properties, it serves as an ideal energy conversion system for GCMRs [3,4].

GCMR Brayton cycles typically employ supercritical carbon dioxide ($S\text{-}CO_2$), helium, or helium-xenon (He-Xe) as the coolant. Several scholars have conducted research on thermodynamic performance of GCMR Brayton cycle using conventional working fluids such as $S\text{-}CO_2$ and helium. For example, Ahn et al. [5] conducted a comparative study on the power conversion efficiency and physical size of small modular reactors Brayton cycles using $S\text{-}CO_2$, He, and nitrogen as working fluids. By considering turbomachinery design and system volume constraints, they proposed new layouts for these three working fluids. Bae et al. [6] investigated the effect of component performance and system size on the thermodynamic performance and operating conditions of helium and $S\text{-}CO_2$ Brayton cycles in small high temperature gas-cooled reactors. The results indicated that the $S\text{-}CO_2$ cycle could be a promising alternative for very small-scale systems. Kim et al. [7] conducted an optimization design of the cycle scheme and key components by considering the constraints of the thermodynamic performance, system volume, and weight of the small modular reactor $S\text{-}CO_2$ Brayton cycle. Li et al. [8] employed optimization algorithms to optimize key parameters of various small modular lead-cooled fast reactor $S\text{-}CO_2$ Brayton cycles with the objectives of maximizing thermoelectric conversion efficiency and minimizing electricity production costs simultaneously. The results

* Corresponding author.
E-mail address: xy-yang@tsinghua.edu.cn (X. Yang).

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Nomenclature

B	second virial coefficient
C	third virial coefficient
C_p	specific heat of constant pressure, [J • kg^{-1} • K^{-1}]
C_V	specific heat of constant volume, [J • kg^{-1} • K^{-1}]
G	Mass flow rate, [kg • s^{-1}]
M	molar mass, [kg • mol^{-1}]
p	pressure, [Pa]
q	specific power, [kJ • kg^{-1}]
R	gas constant, [J • mol^{-1} • K^{-1}]
w	specific power of the turbomachine, [kJ • kg^{-1}]
T	temperature, [K]
V	volume of the component, [m^3]
V^*	critical molar volume
$\dot{V}$	volumetric flow rate, [m^3 • s^{-1}]
x_{He}	helium fraction in helium-xenon
Z	compression factor

Abbreviations

GCMR	gas-cooled micro-reactor
He-Xe	helium-xenon mixture

Greek letters

α	recuperator effectiveness
φ	adiabatic coefficient
ξ	pressure loss coefficient of the component
γ	pressure ratio of the turbomachine
η	efficiency of components
ρ	density, [kg • m^{-3}]
τ	temperature ratio
θ	relative temperature

Subscripts

Coo	cooler
Com	compressor
Gen	generator
Mec	mechanical
Tur	turbine
Rea	reactor
Rec	recuperator

indicated that the recompression cycle is the optimal to operate with more working conditions. Sondelski et al. [9] investigated the effect of key parameters on the cycle performance of the S-CO_2 Brayton cycle for space gas-cooled reactors to meet power requirements and mass constraints. The design solution that minimizes the system's mass was obtained through multi-objective optimization. Rao et al. [10] conducted multi-objective optimization of the recompression S-CO_2 Brayton cycle, taking into account recuperator design parameters, and obtained the Pareto frontier by the genetic algorithm. Yao et al. [11] introduced a one-dimensional design approach for S-CO_2 Brayton cycles, emphasizing the integration of system conceptual design with preliminary component design.

The aforementioned studies indicate that, while ensuring the miniaturization of turbomachinery and heat exchange components, the S-CO_2 GCMR Brayton cycle exhibits high efficiency and good thermodynamic performance at moderate turbine inlet temperatures. However, research by [12,13] has shown that nickel-based alloys experience accelerated oxidation and carburization in high-temperature and high-pressure CO_2 environments, which exacerbates components corrosion. Therefore, it is necessary to select more stable gases as the coolant.

Helium, with its excellent chemical stability and thermal conductivity, is commonly utilized as the working fluid in Brayton cycles for high-temperature gas-cooled reactors [14]. However, helium turbomachinery faces significant aerodynamic loads, resulting in larger numbers of stages and increased volume. The introduction of xenon into helium enhances the heat transfer characteristic of the He-Xe, significantly reduces aerodynamic loads and the number of stages in the turbomachinery, and greatly minimizes the system's volume and weight [15]. Consequently, He-Xe is considered one of the optimal working fluids for GCMR Brayton cycles [16–18].

The GCMR He-Xe Brayton cycle exhibits significant advantages in compactness. As a result, several researchers have explored its application in areas such as deep space exploration and terrestrial mobile micro-reactors. For example, Fuller et al. [19] conducted a comparative study on the performance of space fission reactor Brayton cycles using He-Xe, S-CO_2, and argon as working fluids, finding that the He-Xe working fluid exhibited the highest cycle efficiency and the smallest heat exchanger volume. El-Genk et al. [20–22] investigated the effect of He-Xe composition on system thermodynamic performance and found that the He-Xe working fluid increases pressure loss, leading to a decrease in Brayton cycle efficiency compared to helium working fluid. Xu et al. [23] explored the effects of real gas properties of He-Xe on the power conversion performance of the GCMR Brayton cycel, discovering that using a real gas property calculation model for He-Xe can improve the simulation accuracy of thermodynamic performance. These studies did not account for constraints related to system size and weight. Consequently, several studies have focused on multi-objective optimization targeting power conversion efficiency, volume, and weight. Harnack et al. [24] assessed the sensitivity effect of component performance to the power conversion performance and weight of space reactor He-Xe Brayton cycles. Ribeiro et al. [25,26] examined the effect of heat exchanger size on the thermodynamic performance of space reactor He-Xe Brayton cycles and proposed an optimization method integrating heat exchanger volume and system efficiency. Liu et al. [27] conducted dual-objective optimization studies aimed at maximizing the efficiency while minimizing the weight of space reactor He-Xe Brayton cycles. Ma et al. [15,28] investigated the effects of noble gas binary mixtures on thermal efficiency and weight of the space reactor Brayton cycle and discovered He-Xe is the ideal coolant due to its ability to deliver the highest efficiency and lowest weight. Guan et al. [29,30] investigated the thermo-economic performance of terrestrial micro-reactors Brayton cycles using different coolants, including helium, He-Xe, and S-CO_2, analyzed the effect mechanisms of cycle parameters on the power conversion performance of the system, and obtained the optimal thermal solutions for the system.

Previous studies have analyzed the effects of cycle parameters on the power conversion performance of the GCMR Brayton cycle system and optimized the thermal design to achieve the best performance. However, these studies have not considered the constraints of engineering limitations such as reactor pressure vessel temperature. The power conversion performance and optimal thermal design under reactor pressure vessel temperature limitations remain unclear.

To address existing research gaps and enhance the study of GCMR Brayton cycle design and optimization with He-Xe as the working fluid, this research developed a mathematical model of the cycle examined how key parameters affect its thermal performance. Additionally, by taking the temperature of the reactor pressure vessel into account, the optimal design scheme was identified by maximizing cycle efficiency within its temperature constraints.

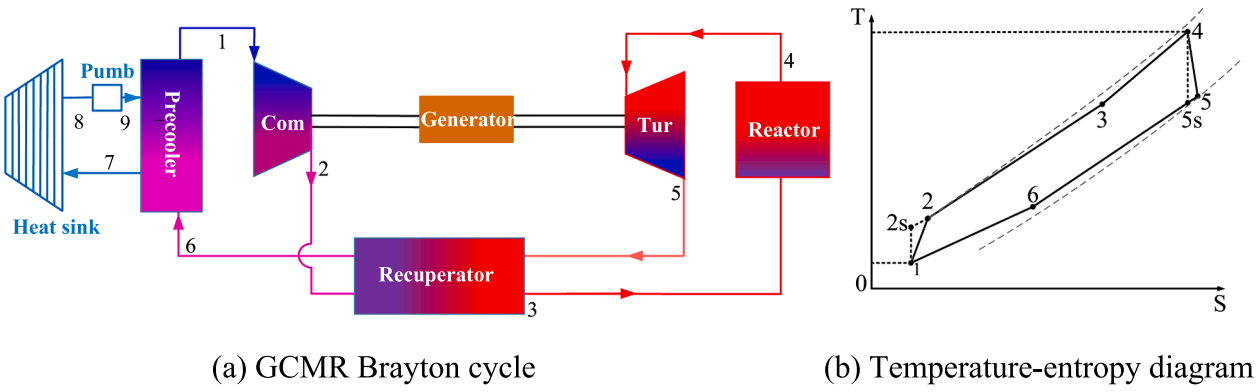


Fig. 1. Schematic diagram of the GCMR Brayton cycle.

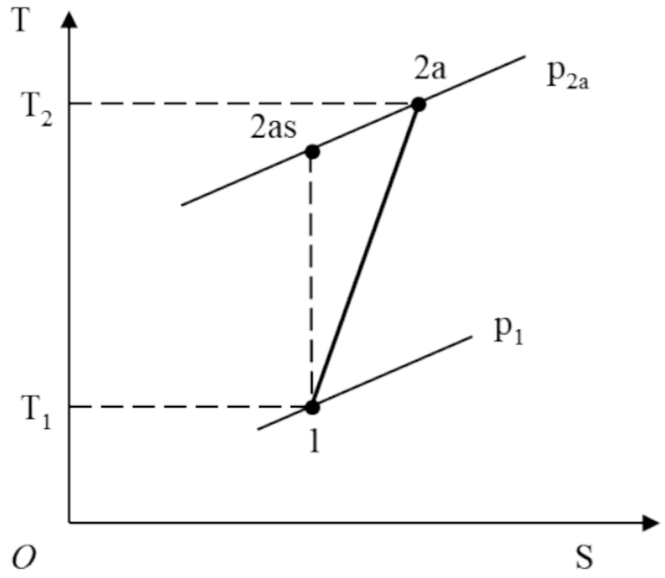


Fig. 2. Compression process of compressor.

2. System description

The GCMR Brayton cycle utilizes the reactor as a heat source and He-Xe as the working fluid, as shown in Fig. 1(a). The He-Xe exits the reactor at node 4 and enters the turbine, where it expands to generate power (process 4 → 5). The high-temperature exhaust from the turbine then flows through the heat rejection side of the recuperator to release residual heat (process 5 → 6) and subsequently through the heat rejection side of the pre-cooler to dissipate waste heat (process 6 → 1). Afterward, the low-temperature gas from the pre-cooler is compressed in the compressor to increase its pressure (process 1 → 2), after which it moves through the heat absorption side of the recuperator to recover heat (process 2 → 3). The He-Xe then returns to the reactor to absorb heat (process 7 → 4), completing a closed-loop cycle. The heat dissipation loop employs water or liquid sodium as the coolant. The fluid absorbs heat on the heat absorption side of the pre-cooler (process 9 → 7), then flows into the heat sink to dissipate heat into the environment (process 7 → 8), and finally, after being pressurized by the pump, returns to the pre-cooler (process 8 → 9). The system's temperature-entropy diagram is presented in Fig. 1(b).

3. Mathematical model and validation

In this section, a mathematical model for the GCMR Brayton cycle is established using the conservation equations for mass, momentum, and energy for each component. The model is based on the following assumptions [31,32]:

(1) The processes in the GCMR Brayton cycle are steady state;
(2) Heat loss of the GCMR Brayton cycle is neglected;
(3) Pressure drops across components are assumed to be constant.

3.1. Compression process of the compressor

Fig. 2 shows the T-S diagram of the compressor compression process. The initial parameters of the process are pressure p_1 and temperature T_1, while the final parameters are pressure p_2 and temperature T_2. The process 1 → 2 s and 1 → 2 represent the isentropic compression and actual compression processes.

The compressor pressure ratio γ_{Com} is defined as the ratio between the compressor outlet pressure p_2 and the inlet pressure p_1, and it is calculated as follows:

$$\gamma_{\mathrm{Com}} = \frac{p_2}{p_1} \tag{1}$$

The compressor efficiency η_{Com} is defined as the ratio of the work required for an ideal compression process to the work used in the actual compression process:

$$\eta_{\mathrm{Com}} = \frac{C_{p,1-2s}(T_{2s} - T_1)}{C_{p,1-2}(T_2 - T_1)} \tag{2}$$

where $C_{p,1-2s}$ and $C_{p,1-2}$ are specific heat capacities at constant pressure for the isentropic compression and actual compression processes, respectively.

The process 1 → 2 s is an adiabatic isentropic process. The equation of state for real gases and adiabatic equation can be expressed as:

$$pV = ZRT \tag{3}$$

$$pV^k = const \tag{4}$$

where Z, k and R are the compressibility factor, the ratio of specific heats, and the gas constant of the working fluid, respectively.

According to Eqs. (3) and (4), the temperature and pressure in the isentropic compression process satisfy the following relationship:

$$\frac{T_{2s}}{T_1} = \left(\frac{p_2}{p_1}\right)^{\varphi_{1-2s}} \tag{5}$$

where φ_{1-2s} is the adiabatic index, which can be expressed as: $\varphi_{1-2s} = \frac{k-1}{k}$; T_{2s} is the outlet temperature of the isentropic compression process.

According to Eqs. (1), (2) and (5), the compressor outlet temperature and specific work can be expressed as follows (The detailed derivation process is provided in Appendix A):

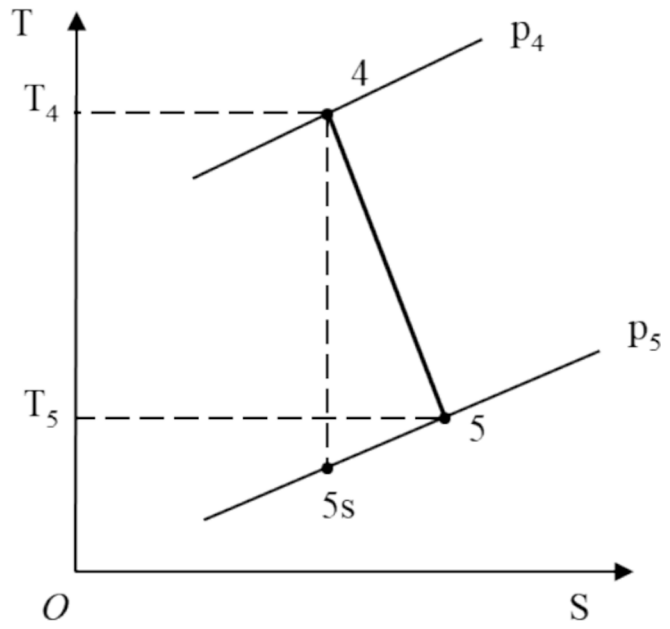


Fig. 3. Expansion process of turbine.

$$T_2 = T_4\tau^{-1}\left[1+\frac{C_{p,1-2s}}{C_{p,1-2}}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right)\right] \tag{6}$$

$$w_{\mathrm{Com}} = T_4\tau^{-1}C_{p,1-2s}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right) \tag{7}$$

where T_4 represents the turbine inlet temperature, and τ is the cycle temperature ratio, defined as $\tau = T_4/T_1$.

3.2. Expansion process of the turbine

Fig. 3 shows the T-S diagram of the actual expansion process in the turbine. The initial parameters of the process are pressure p_4 and temperature T_4, while the final parameters are pressure p_5 and temperature T_5. In both the turbine and compressor, heat exchange with the surroundings is minimal and can be considered negligible. Therefore, the expansion process of the gas in the turbine can be approximated as adiabatic. The ideal expansion process 4 → 5 s in the turbine is a reversible adiabatic process, or isentropic expansion. The actual expansion process 4 → 5 is an irreversible adiabatic process.

For the pressure changes during the turbine expansion process, the turbine expansion ratio γ_{Tur} is calculated as follows:

$$\gamma_{\mathrm{Tur}} = \frac{p_4}{p_5} \tag{8}$$

According to the pressure balance in a closed-loop cycle, the turbine inlet and outlet pressures satisfy the following relationship:

$$p_4 = p_2(1-\xi_1-\xi_2) \tag{9}$$

$$p_5 = p_1 + p_2(\xi_3+\xi_4) \tag{10}$$

where ξ_1, ξ_2, ξ_3, ξ_4 represent the pressure loss coefficients for the working fluid as it passes through the heat absorption side of the recuperator, the reactor, the heat rejection side of the recuperator, and the heat rejection side of the pre-cooler. These coefficients are defined as the ratio of pressure loss to the compressor outlet pressure.

By solving the above pressure relationships simultaneously, the turbine expansion ratio is obtained as follows:

$$\gamma_{\mathrm{Tur}} = \frac{(1-\xi_1-\xi_2)\gamma_{\mathrm{Com}}}{1+(\xi_3+\xi_4)\gamma_{\mathrm{Com}}} \tag{11}$$

The turbine efficiency η_{Tur} is the ratio of the work done during the actual expansion process to the work done during the ideal expansion process:

$$\eta_{\mathrm{Tur}} = \frac{C_{p,4-5}(T_4-T_5)}{C_{p,4-5s}(T_4-T_{5s})} \tag{12}$$

where $C_{p,4-5s}$ and $C_{p,4-5}$ are specific heat capacities at constant pressure for the isentropic expansion and actual expansion processes, respectively.

For an isentropic process, the following relationship holds:

$$\frac{T_4}{T_{5s}} = \left(\frac{p_4}{p_5}\right)^{\varphi_{4-5s}} \tag{13}$$

where φ_{4-5s} is the adiabatic index, which can be expressed as: $\varphi_{4-5s} = \frac{k-1}{k}$; T_{5s} is the outlet temperature of the isentropic expansion process.

According to Eq. (8) through (13), the turbine outlet temperature and the expansion work can be expressed as follows:

$$T_5 = T_4\left[1-\frac{C_{p,4-5s}}{C_{p,4-5}}\eta_{\mathrm{Tur}}\left(1-\gamma_{\mathrm{Tur}}^{-\varphi_{4-5s}}\right)\right] \tag{14}$$

$$w_{\mathrm{Tur}} = T_4C_{p,4-5s}\eta_{\mathrm{Tur}}\left(1-\gamma_{\mathrm{Tur}}^{-\varphi_{4-5s}}\right) \tag{15}$$

3.3. Heat recovery process of the recuperator

The gas, pressurized by the compressor, enters the heat absorption side of the recuperator to absorb the residual heat from the turbine exhaust. According to the heat balance of the heat exchanger, the specific power q_{Rec} of the recuperator is calculated as follows:

$$q_{\mathrm{Rec}} = C_{p,2-3}(T_3-T_2) = C_{p,5-6}(T_5-T_6) \tag{16}$$

The recuperation effectiveness α is the ratio of the actual recovered heat to the maximum possible recovered heat, and it can be expressed as:

$$\alpha = \frac{C_{p,2-3}(T_3-T_2)}{C_{p,2-5}(T_5-T_2)} \tag{17}$$

From Eq. (17), the temperature of the working fluid at the outlet of the heat absorption side of the recuperator is:

$$T_3 = \alpha\frac{C_{p,2-5}}{C_{p,2-3}}(T_5-T_2)+T_2 \tag{18}$$

3.4. Cooling process of the pre-cooler

In the pre-cooler, the gas releases heat on one side, while on the other side, the working fluid absorbs heat and then flows through the radiator to dissipate heat into the environment. The specific power q_{Coo} of the pre-cooler is calculated as follows:

$$q_{\mathrm{Coo}} = C_{p,6-1}(T_6-T_1) \tag{19}$$

3.5. Heat absorption process of the reactor

The He-Xe working fluid enters the reactor to absorb heat. Based on the derivation process in Appendix B, the reactor inlet temperature T_3 can be expressed as follow:

$$T_3 = \alpha\frac{C_{p,2-5}}{C_{p,2-3}}T_4\left[1-\frac{C_{p,4-5s}}{C_{p,4-5}}\eta_{\mathrm{Tur}}\left(1-\gamma_{\mathrm{Tur}}^{-\varphi_{4-5s}}\right)\right]+\left(1-\alpha\frac{C_{p,2-5}}{C_{p,2-3}}\right)T_4\tau^{-1}\left[1+\frac{C_{p,1-2s}}{C_{p,1-2}}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right)\right] \tag{20}$$

The specific power q_{Rea} of the reactor can be expressed as the difference between the specific enthalpies at the reactor outlet and inlet:

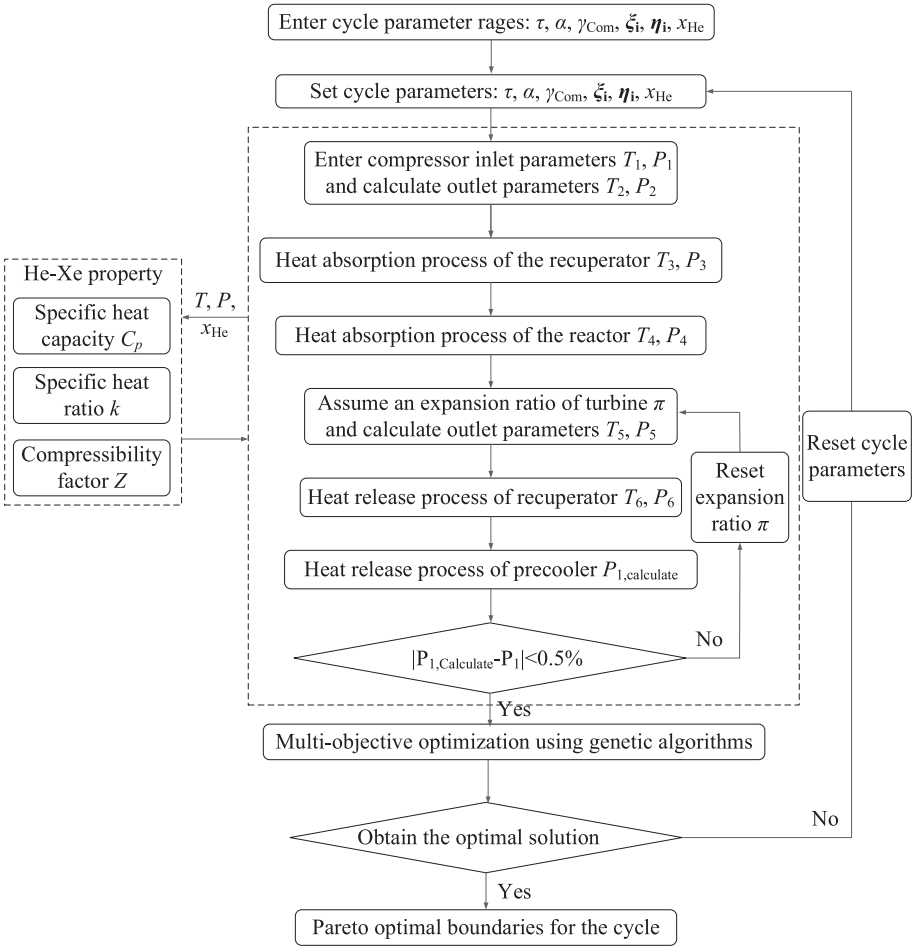


Fig. 4. Analysis and optimization program of the GCMR Brayton cycle.

$$q_{\mathrm{Rea}} = C_{\mathrm{p},3-4}T_4\left\{1-\alpha\frac{C_{p,2-5}}{C_{p,2-3}}\left[1-\frac{C_{p,4-5s}}{C_{p,4-5}}\eta_{\mathrm{Tur}}\left(1-\gamma_{\mathrm{Tur}}^{-\varphi_{4-5s}}\right)\right]-\left(1-\alpha\frac{C_{p,2-5}}{C_{p,2-3}}\right)\tau^{-1}\left[1+\frac{C_{p,1-2s}}{C_{p,1-2}}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right)\right]\right\} \tag{21}$$

3.6. Cycle efficiency and specific power generation

The specific power generation of the cycle w_{e} is expressed as:

$$w_{\mathrm{e}} = \eta_{\mathrm{Mec}}\eta_{\mathrm{Gen}}\left(w_{\mathrm{Tur}}-w_{\mathrm{Com}}\right) = T_4\eta_{\mathrm{Mec}}\eta_{\mathrm{Gen}}\left\{C_{p,4-5s}\eta_{\mathrm{Tur}}\left(1-\gamma_{\mathrm{Tur}}^{-\varphi_{4-5s}}\right)-\tau^{-1}C_{p,1-2s}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right)\right\} \tag{22}$$

$$\eta_{\mathrm{Mec}} = 1-\frac{w_{\mathrm{Loss}}}{w_{\mathrm{Tur}}-w_{\mathrm{Com}}} \tag{23}$$

where, w_{Loss} represents the shaft losses. According to the study in reference [22], the shaft losses w_{Loss} include the thrust bearing losses w_{thrust}, the windage losses of the turbine and compressor disks $w_{\mathrm{Disk}}^{\mathrm{Tur}}$ and $w_{\mathrm{Disk}}^{\mathrm{Com}}$, and the mechanical losses of the shaft w_{shaft}. This study focuses on the thermodynamic performance of the cycle; therefore, to simplify the model, the shaft mechanical efficiency is assumed to be 0.98.

The cycle efficiency η_{e} is defined as the ratio of the output power to the reactor power, and can be represented as:

$$\eta_{\mathrm{e}} = \frac{w_{\mathrm{e}}}{q_{\mathrm{Rea}}} = \frac{\eta_{\mathrm{Mec}}\eta_{\mathrm{Gen}}\left\{\frac{C_{p,4-5s}}{C_{\mathrm{p},3-4}}\eta_{\mathrm{Tur}}\left(1-\gamma_{\mathrm{Tur}}^{-\varphi_{4-5s}}\right)-\frac{C_{p,1-2s}}{C_{\mathrm{p},3-4}}\tau^{-1}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right)\right\}}{1-\alpha\frac{C_{p,2-5}}{C_{p,2-3}}\left[1-\frac{C_{p,4-5s}}{C_{p,4-5}}\eta_{\mathrm{Tur}}\left(1-\gamma_{\mathrm{Tur}}^{-\varphi_{4-5s}}\right)\right]-\left(1-\alpha\frac{C_{p,2-5}}{C_{p,2-3}}\right)\tau^{-1}\left[1+\frac{C_{p,1-2s}}{C_{p,1-2}}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right)\right]} \tag{23}$$

where η_{Mec} and η_{Gen} are mechanical efficiency of the shaft and generator efficiency. The shaft mechanical efficiency η_{Mec} can be defined as [22]:

In Eq. (23), the specific heat capacity C_p and the adiabatic index φ are functions of the fraction of helium in the He-Xe working fluid x_{He}.

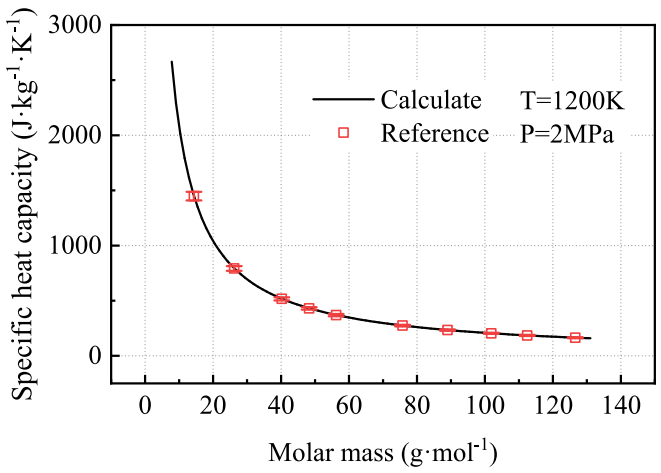


Fig. 5. Comparison of physical property models.

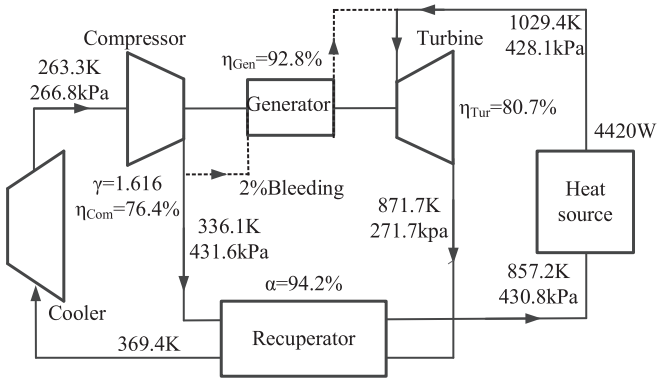


Fig. 6. Schematic diagram of the BIPS experiment.

Table 1
The verification results of the model and program.

Parameters	Pressure (kPa)			Temperature (K)		
	BIPS	Calculate	Error	BIPS	Calculate	Error
Turbine inlet	425.4	425.4	0.00 %	1025.0	1018	−0.69 %
Turbine outlet	271.7	277.7	2.16 %	871.7	884	1.39 %
Cooler inlet	–	274.3	–	369.4	366.8	−0.71 %
Compressor inlet	266.8	267.1	0.11 %	263.3	263.3	0.00 %
Compressor outlet	431.6	431.6	0.00 %	336.1	342.3	1.81 %
Heat source inlet	430.8	430.8	0.00 %	857.2	860.9	0.43 %
Heat source outlet	428.1	428.1	0.00 %	1029.4	1029.4	0.00 %
Cycle efficiency	27.2 %	27.8 %	2.16 %			

Therefore, the cycle efficiency can be further expressed as the following function:

$$\eta_e = \eta_e(\tau, \alpha, \gamma_{Com}, \boldsymbol{\xi_i}, \boldsymbol{\eta_i}, x_{He}) \tag{24}$$

$$\boldsymbol{\xi_i} = [\xi_1, \xi_2, \xi_3, \xi_4]^T \tag{25}$$

$$\boldsymbol{\eta_i} = [\eta_{Tur}, \eta_{Com}, \eta_{Mec}, \eta_{Gen}]^T \tag{26}$$

From Eq. (24), it can be seen that the efficiency of the GCMR He-Xe Brayton cycle is determined by the system parameters $\tau, \alpha, \gamma_{Com}, \boldsymbol{\xi_i}, \boldsymbol{\eta_i}$, and the working fluid parameters x_{He}.

Table 2
Initial parameter values.

Parameter	Symbol	Unit	Value
Compressor outlet pressure	P_2	MPa	2
Turbine inlet temperature	T_4	K	1073
Compressor inlet temperature	T_1	K	303
Pressure ratio	γ_{Com}	–	2
Recuperator effectiveness	α	%	90
Compressor efficiency	η_{Com}	%	87.5
Turbine efficiency	η_{Tur}	%	89.5
Mechanical efficiency	η_{Mec}	%	95
Generator efficiency	η_{Gen}	%	98

3.7. Physical property model of He-Xe working fluid

Due to the deviation of high molecular mass gases from ideal gas, this study treats the He-Xe working fluid as a real gas in all calculations of physical parameters. Based on the property models for inert gases established by Xu et al. [23]and El-Genk et al. [33], the density, specific heat capacity, and compressibility factor of the He-Xe are computed using second and third virial coefficients, as shown in Appendix C.

3.8. Program and model validation

The analysis and optimization program for the GCMR Brayton cycle is developed and shown in Fig. 4. First, based on engineering and empirical constraints, the system parameters $\tau, \alpha, \gamma, \boldsymbol{\xi_i}, \boldsymbol{\eta_i}$ and the working fluid parameter x_{He} are selected within their respective ranges. A set of parameters is set as input for the program, and in combination with the He-Xe properties calculation module, subroutines for the compressor, recuperator, reactor, turbine, and gas cooler are sequentially called by solving the mass, momentum, and energy conservation equations for each component. The convergence of the compressor inlet pressure is evaluated to ensure pressure balance in the closed cycle. Finally, the node thermodynamic parameters and the system performance are determined. By varying the input parameters within their respective ranges, the influence of key parameters on the cycle's thermodynamic performance is analyzed. On this basis, the thermodynamic performance results under different input parameter sets are fed into a genetic algorithm for multi-objective optimization, to maximize cycle efficiency and minimize reactor inlet temperature. If the optimization result does not reach the Pareto optimal frontier, the input parameters are reset, and the thermodynamic performance is recalculated until the Pareto optimal frontier is achieved.

3.8.1. Validation of physical property model

The thermodynamic performance calculations of the system require the specific heat capacity, heat capacity ratio, and other physical parameters of the He-Xe working fluid. This study computed the specific heat capacity of the He-Xe at 1200 K and 2 MPa as a function of the molar mass of the He-Xe. These results were compared with some data reported in the literature [34], as illustrated in Fig. 5. The maximum relative error between the calculated values and the reference values is 2.7 %. The results align well with the literature data, confirming the accuracy of the physical property calculation model.

3.8.2. Thermodynamic model validation

Experimental research on the He-Xe Brayton cycle is relatively limited. In the 1970s, the United States initiated the Brayton Isotope Power System (BIPS) project, which conducted experimental studies on the He-Xe Brayton cycle as illustrated in Fig. 6 [35,36]. Consequently, this study uses the experimental data from BIPS to validate the accuracy of the model and program. By inputting initial parameters such as

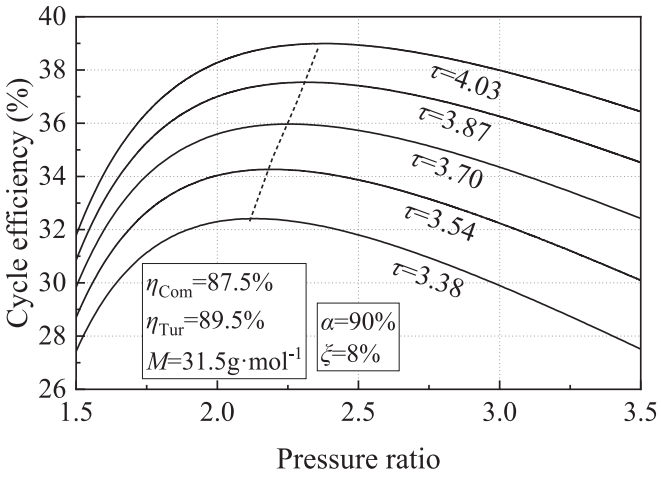


Fig. 7. The relationship between cycle efficiency, pressure ratio and temperature ratio.

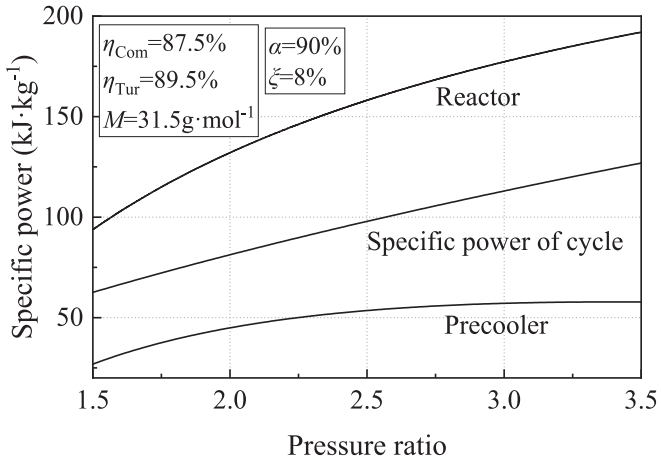


Fig. 8. Variation of cycle specific power with pressure ratio.

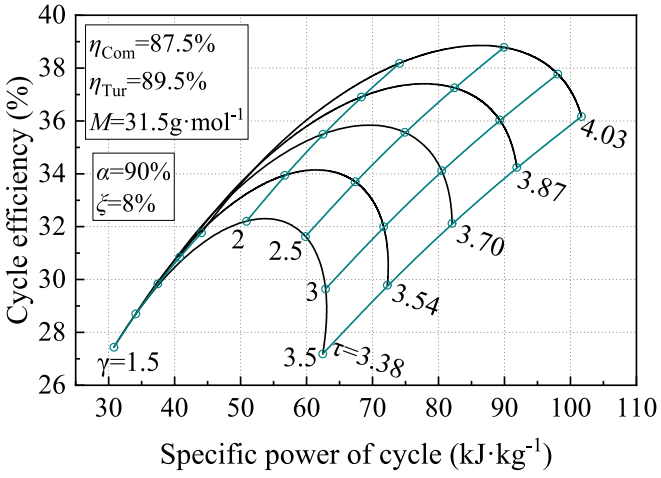


Fig. 9. The relationship between cycle efficiency, specific work, temperature ratio, and pressure ratio.

temperature ratio, recuperator effectiveness, pressure ratio, and turbomachine efficiencies into the model in this study, the results are as shown in Table 1. The maximum errors for node temperatures, pressures and power generation efficiency are 1.81 %, 2.16 % and 2.16 %,

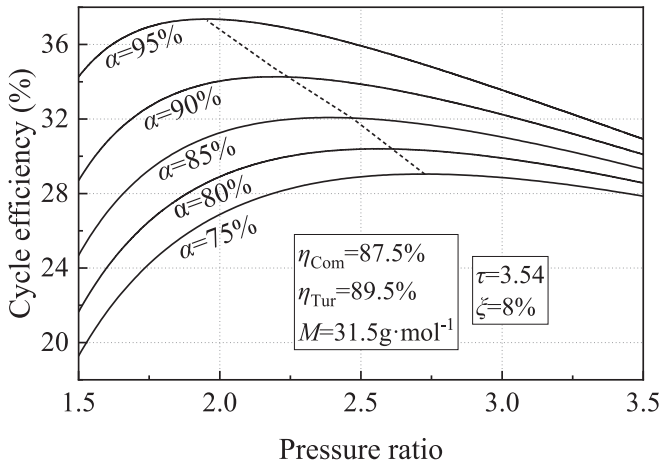


Fig. 10. The relationship between cycle efficiency, pressure ratio and recuperator effectiveness.

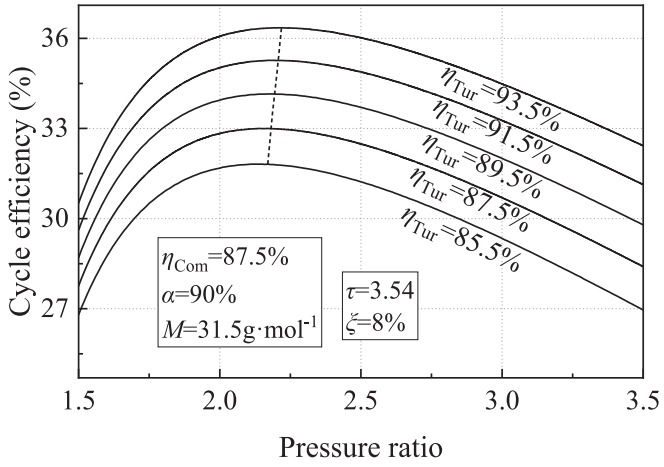


Fig. 11. The relationship between cycle efficiency, pressure ratio and turbine efficiency.

respectively. The model calculations align well with the test data, thereby confirming the accuracy of the model and the program.

4. Results analysis and discussion

The variations in temperature ratio, component efficiency, and so on alter the performance of key components, which changes the power conversion performance of the cycle. Changes in the He-Xe composition result in variations in properties such as specific heat, adiabatic coefficient, causing fluctuations in the cycle node parameters, thus affecting the cycle thermal performance. Therefore, this section examines how key parameters influence the power conversion performance of the system. The initial parameters are specified in Table 2 [37].

4.1. Influence of key parameters on the thermodynamic performance of the cycle

Analysis of the expression for cycle efficiency reveals that a higher temperature ratio results in higher cycle efficiency. Therefore, cycle efficiency is a monotonically increasing function of the temperature ratio. The relationship between cycle efficiency and the temperature ratio is shown in Fig. 7. Under constant conditions, increasing the turbine inlet temperature or decreasing the compressor inlet temperature to raise the temperature ratio will result in an increase in turbine output

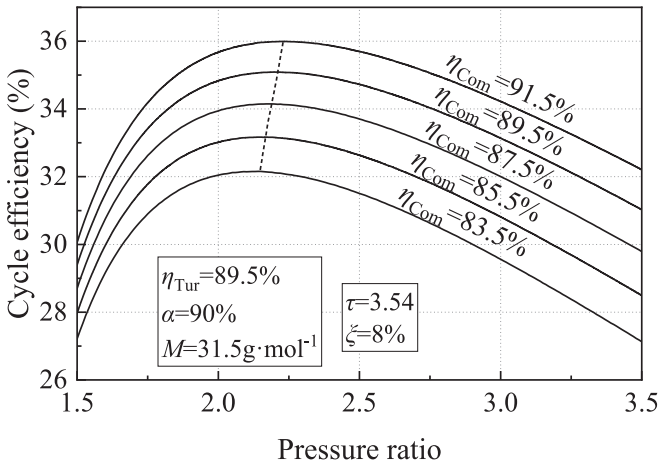


Fig. 12. The relationship between cycle efficiency, pressure ratio and compressor efficiency.

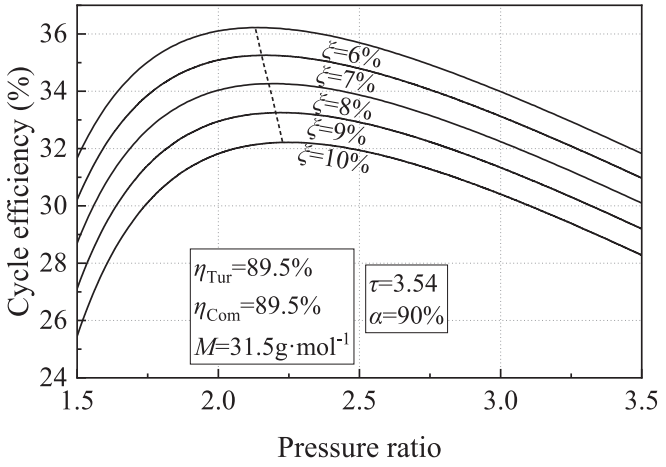


Fig. 13. The relationship between cycle efficiency, pressure ratio and pressure loss coefficient.

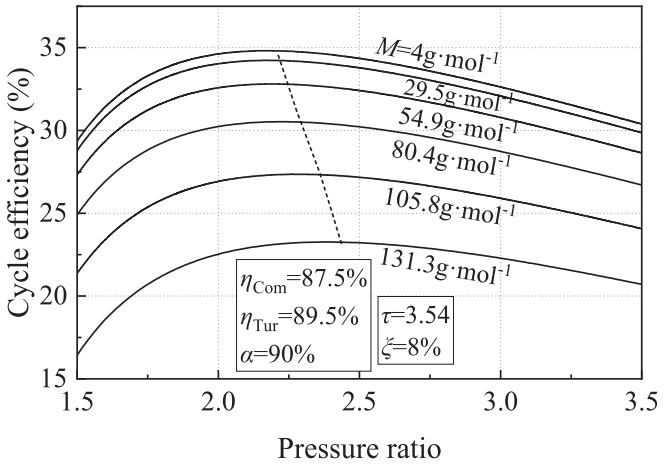


Fig. 14. The relationship between cycle efficiency, pressure ratio and He-Xe molar mass.

specific power or a reduction in compressor specific power, leading to an increase in the specific power of the cycle and, consequently, an increase in cycle efficiency.

From Fig. 7, for a given temperature ratio, the cycle efficiency is a

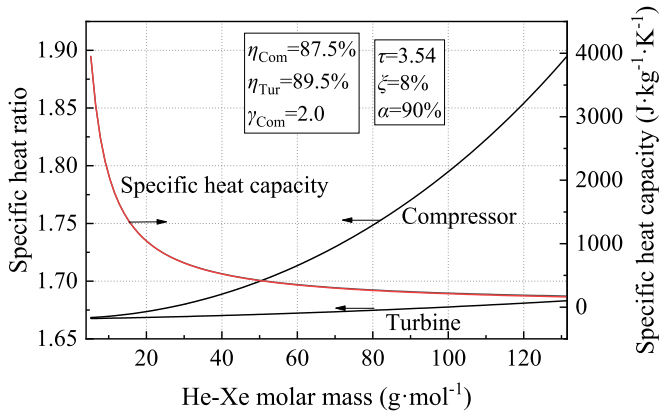


Fig. 15. The relationship between properties and He-Xe molar mass.

non-monotonic function of the pressure ratio. This can be explained by analyzing the cycle specific power, pre-cooler specific power, and reactor specific power. Fig. 8 shows the variation of cycle specific power, pre-cooler specific power, and reactor specific power with the pressure ratio. As the pressure ratio increases from a relatively low value, the cycle's specific power rises monotonically with the pressure ratio, which helps enhance the cycle's efficiency. However, the heat rejection in the pre-cooler and the heat absorption from the reactor also increase simultaneously, which is detrimental to cycle efficiency. Due to these opposing factors, the relationship between cycle efficiency and pressure ratio is not monotonic and exhibits optimal value. As the cycle temperature ratio increases from 3.38 to 4.03, the maximum cycle efficiency and the corresponding pressure ratio increase from 32.31 % and 2.11 to 38.86 % and 2.34, respectively.

The combined effects of pressure ratio and temperature ratio on cycle specific power and efficiency are illustrated in Fig. 9. As shown in the figure, for a given cycle pressure ratio, the cycle specific power increases with an increase in temperature ratio. Within a certain range of pressure ratios, for a given temperature ratio, the cycle specific power also increases with an increase in pressure ratio.

Recuperator effectiveness, pressure loss coefficient, and turbomachine efficiency directly affect the cycle performance. Figs. 10 to 13 illustrate the influence of recuperator effectiveness, turbomachine efficiencies, and total pressure loss coefficient on cycle efficiency. Cycle efficiency is a monotonically increasing function of turbine efficiency and compressor efficiency, and a monotonically decreasing function of pressure loss coefficient.

The recuperator effectively harnesses the residual heat from the turbine outlet, enabling higher cycle efficiency at lower cycle pressure ratios. Therefore, recuperator effectiveness is a key factor influencing the thermdynamic performance of the system. As the recuperator effectiveness increases, the heat recovery by the recuperator increases, leading to a decrease in the heat absorbed by the He-Xe in the reactor, and an increase in cycle efficiency, as shown in Fig. 10. When the recuperator effectiveness increases from 75 % to 95 %, the maximum cycle efficiency rises from 28.83 % to 37.31 %, while the corresponding pressure ratio decreases from 2.7 to 1.94.

Turbomachine efficiencies are key parameters influencing the power conversion performance of the cycle. An increase in turbine or compressor efficiency leads to an increase in turbine output power or a decrease in compressor power, thereby enhancing specific power and efficiency of the cycle, as shown in Figs. 11 and 12. When turbine efficiency increases from 85.5 % to 93.5 %, cycle efficiency and the corresponding pressure ratio increase from 31.81 % and 2.15 to 36.35 % and 2.22, respectively. Similarly, when compressor efficiency increases from 83.5 % to 91.5 %, cycle efficiency and the corresponding pressure ratio rise from 32.16 % and 2.14 to 35.99 % and 2.22, respectively.

The pressure loss coefficients depend on the flow performance of the

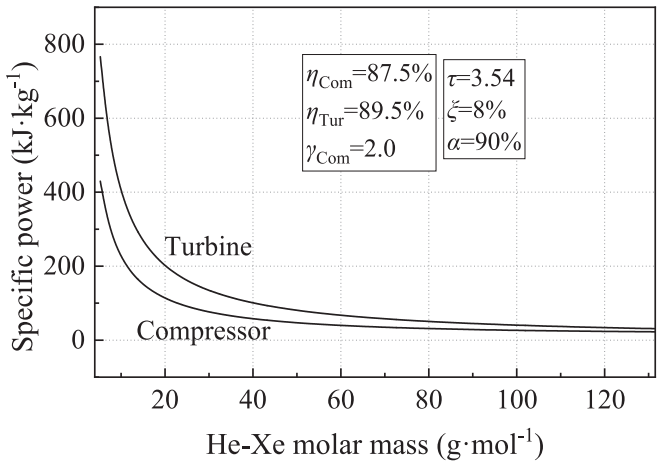


Fig. 16. The relationship between turbomachine specific work and He-Xe molar mass.

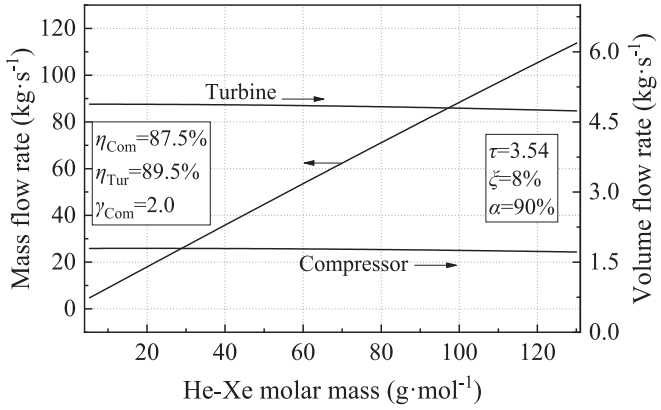


Fig. 17. The relationship between flow rate and He-Xe molar mass.

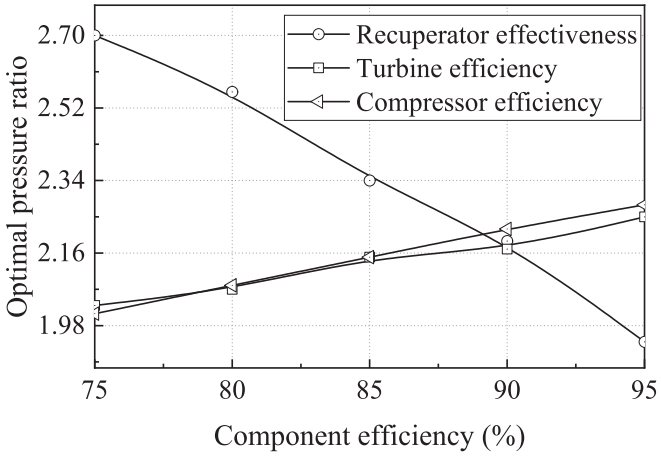


Fig. 18. Variation of optimal pressure ratio with component efficiency.

various components. When the pressure ratio and other parameters are fixed, an increase in pressure loss coefficients in components reduce the pressure drop of the He-Xe across the turbine in a closed-loop system. This leads to a decrease in turbine expansion ratio and output power, resulting in lower cycle specific work and efficiency. As the total pressure loss coefficient increases from 6 % to 10 %, cycle efficiency decreases from 36.11 % to 32.09 %, while the corresponding pressure ratio

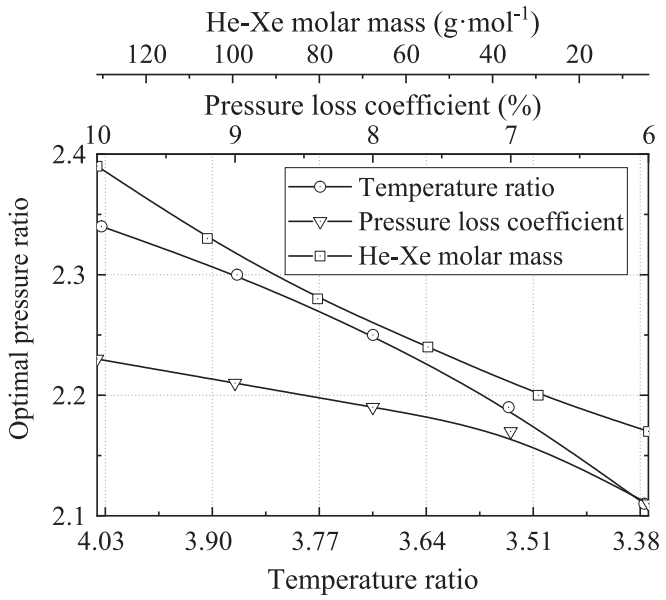


Fig. 19. Variation of optimal pressure ratio with temperature ratio, pressure loss coefficient, and He-Xe molar mass.

increases from 2.11 to 2.23, as shown in Fig. 13.

He-Xe is among the most effective working fluids for Brayton cycles in GCMR. As shown by the expressions for cycle specific power and efficiency, changes in the He-Xe composition lead to variations in physical properties like specific heat and heat capacity ratio, which in turn alter the power conversion performance of the cycle.

With other parameters held constant, the variation of cycle efficiency with He-Xe molar mass and pressure ratio is shown in Fig. 14. It is evident that cycle efficiency is a monotonically decreasing function of the He-Xe molar mass, which can be attributed to the properties of the He-Xe. Fig. 15 illustrates how the specific heat capacity and heat capacity ratio of the He-Xe change with the molar mass. As the He-Xe molar mass increases, the average heat capacity ratio of the He-Xe inside the turbomachinery increases, leading to increased turbomachinery power. However, the significant decrease in the specific heat capacity of the He-Xe results in lower turbomachinery power. Since specific heat capacity is dominant over specific heat capacity ratio in these properties, the specific work of the turbomachinery decreases, as shown in Fig. 16. Due to the lower operating temperature of the compressor, the changes in He-Xe properties are more pronounced. Fig. 15 shows that the heat capacity ratio of the compressor increases more significantly than that of the turbine. The decrease in heat capacity and the significant increase in the heat capacity ratio result in a less pronounced decrease in the compressor specific work compared to the turbine specific work, as shown in Fig. 16. Consequently, the cycle specific work and efficiency decrease. As the He-Xe molar mass increases from 4 $g \bullet mol^{-1}$ to 131.3 $g \bullet mol^{-1}$, the maximum cycle efficiency decreases from 34.82 % to 23.27 %, while the corresponding pressure ratio increases from 2.17 to 2.39.

Volume is a key constraint indicator for the GCMR Brayton cycle, and flow rate is a crucial parameter in component design. Assuming the reactor power is 4 MW, the trends of the mass flow rate and the volume flow rates with the change in the molar mass of He-Xe are shown in Fig. 17. It can be observed that when the molar mass of He-Xe increases, the specific heat capacity of the working fluid decreases, which increases the mass flow rate significantly. The He-Xe density also increases accordingly. Therefore, the volume flow rate of the working fluid remains essentially unchanged, and the radial size of the turbomachinery varies little.

In terms of the volume of the heat exchanger, when the He-Xe molar

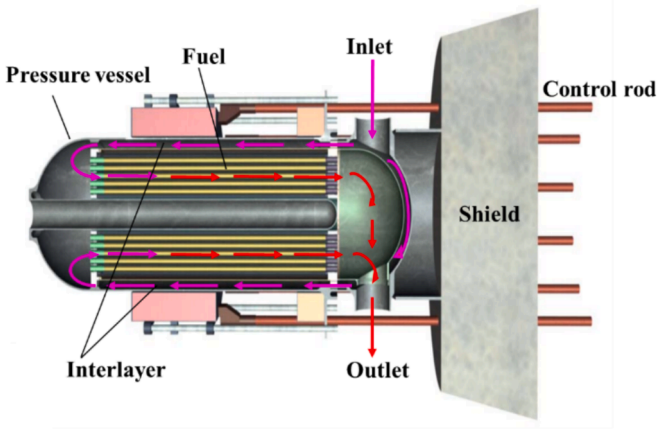


Fig. 20. Schematic diagram of core structure and he-xe working fluid flow path [39].

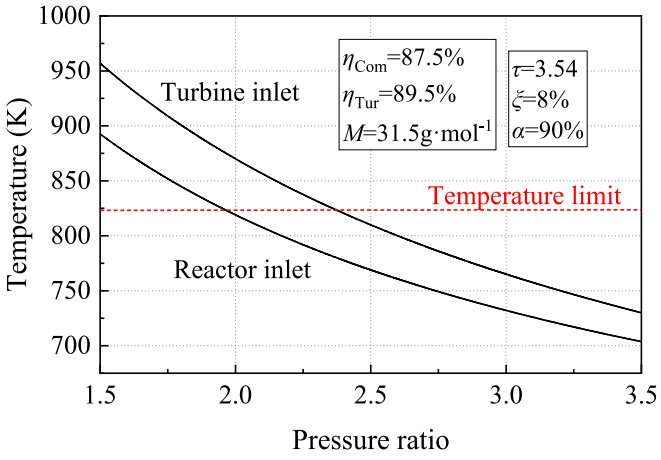


Fig. 21. Variation of reactor inlet temperature with pressure ratio.

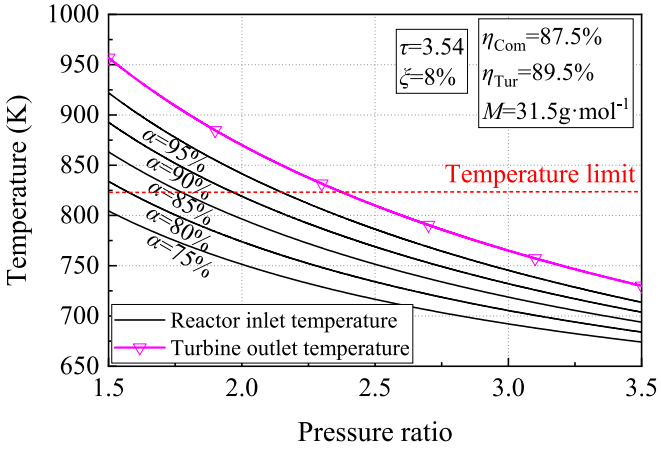


Fig. 22. Effect of pressure ratio on reactor inlet and turbine outlet temperatures at different recuperator effectiveness.

mass increases, the significant increase in the mass flow rate leads to an increase in the pressure loss coefficient. Since the pressure loss coefficients of the heat exchanger are fixed values when analyzing the impact of the He-Xe composition on the thermodynamic performance of the cycle. Therefore, it is necessary to increase the flow cross-sectional area of the heat exchanger to reduce the flow velocity and ensure that

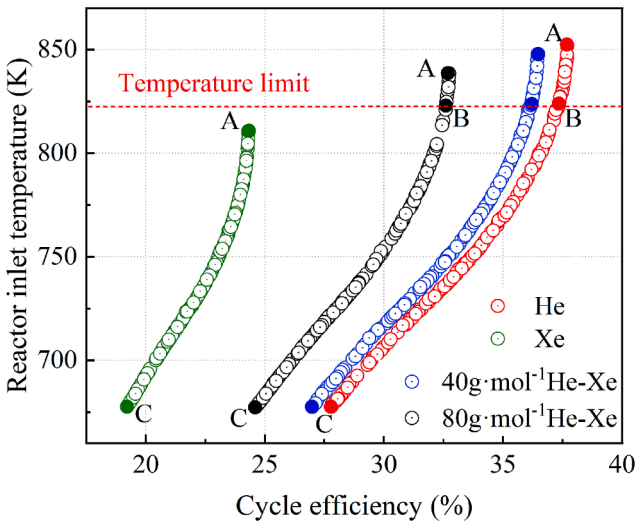


Fig. 23. Pareto optimal boundaries for the cycle.

the pressure loss coefficients remain fixed. Consequently, as the He-Xe molar mass increases, the size of the heat exchanger increases accordingly.

Based on the previous analysis, it is evident that, under constant conditions, cycle efficiency depends solely on the pressure ratio.

$$\eta_e = \eta_e(\gamma) \tag{27}$$

The optimal pressure ratio is the ratio at which the cycle efficiency achieves its peak value. When all other factors in the cycle remain constant, both the optimal pressure ratio and the maximum cycle efficiency are determined.

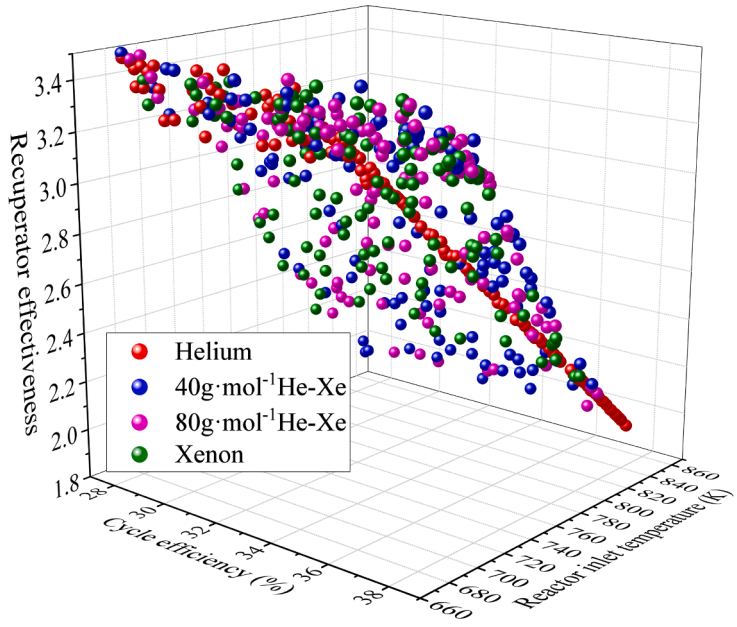
From the above analysis, it is evident that changes in other parameters lead to changes in the optimal pressure ratio. Fig. 18 shows the effects of recuperator effectiveness and turbomachine efficiencies on the optimal pressure ratio. As the recuperator effectiveness increases, the optimal pressure ratio decreases; conversely, as turbine efficiency or compressor efficiency increases, the optimal pressure ratio increases. Fig. 19 illustrates the effect of temperature ratio, pressure loss coefficient, and working fluid composition on the optimal pressure ratio. As the HeXe molar mass, temperature ratio and pressure loss decrease, the optimal pressure ratio decreases.

Compared to other cycle parameters, the optimal pressure ratio is primarily influenced by the recuperator effectiveness. With a turbine efficiency of 89.5 %, a compressor efficiency of 87.5 %, and a total pressure loss coefficient of 8 %, when the helium molar fraction is 78.4 % and the temperature ratio is 3.54, the optimal pressure ratio for a recuperator effectiveness of 75 % is 2.7, whereas for a recuperator effectiveness of 95 %, the optimal pressure ratio is 1.94.

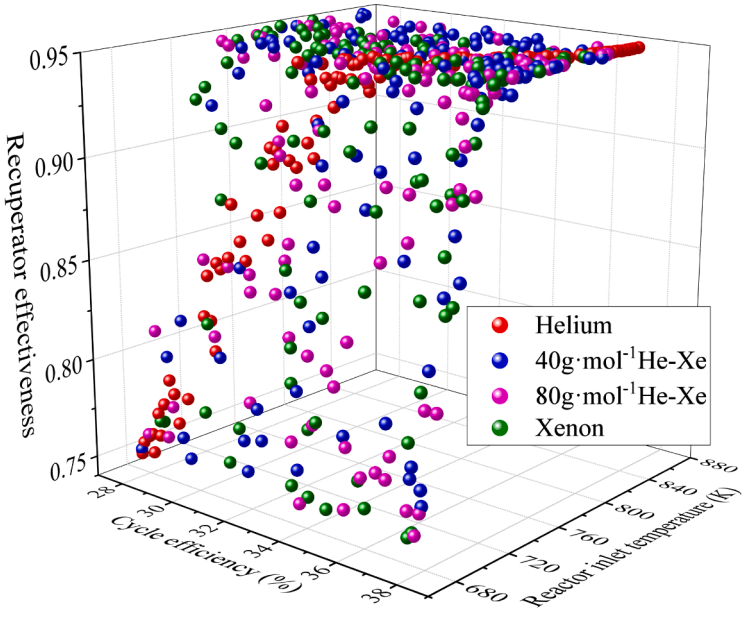
4.2. *Temperature limits of the reactor pressure vessel*

The optimal pressure ratio is accurate for a purely thermodynamic problem, but it is not comprehensive for a practical GCMR Brayton cycle. This is because practical cycles face engineering constraints. For example, the reactor pressure vessel serves as the pressure boundary of the system to prevent the release of radioactivities. To maintain the strength of the pressure vessel, it cannot operate beyond the design temperature. To reduce the temperature of the pressure vessel, a sandwich design, as shown in Fig. 20, is employed in the core of the GCMR. The He-Xe working fluid from the reactor inlet flows through the interlayer surrounding the core and isolates the pressure vessel, which protects the pressure vessel, before entering the core fuel region to absorb heat and exit the reactor.

The choice of reactor inlet temperature directly affects the design of



(a) Pressure ratio



(b) Recuperator effectiveness

Fig. 24. Optimal range of cycle parameters.

the reactor pressure vessel, which further affects the safety and power conversion performance of the system. With current technology, the pressure vessel of a GCMR typically uses SA533 steel with internal gas cooling. Therefore, the reactor inlet temperature limited to 823 K [38]. Although the optimized pressure ratio can achieve maximum cycle efficiency, the required reactor inlet temperature may be too high for practical engineering implementation. Therefore, this section focuses on analyzing the effect of key parameters on the reactor inlet temperature.

From Eq. (20), it is evident that the reactor inlet temperature can also be represented as a function of the system parameters $\tau, \alpha, \gamma, \boldsymbol{\xi_i}, \boldsymbol{\eta_i}$ and the working fluid composition x_{He}:

$$T_3 = T_3(\tau, \alpha, \gamma_{\mathrm{Com}}, \boldsymbol{\xi_i}, \boldsymbol{\eta_i}, x_{\mathrm{He}}) \tag{28}$$

$$\boldsymbol{\xi_i} = [\xi_1, \xi_2, \xi_3, \xi_4]^T \tag{29}$$

Table 3
Optimal design parameters for the helium Brayton cycle.

Parameter	Symbol	Unit	A	B	C
Cycle efficiency	η_e	%	37.70	37.32	27.80
Reactor inlet temperature	T_3	K	851.92	822.60	677.86
Pressure ratio	γ_{Com}	–	1.94	2.18	3.5
Recuperator effectiveness	α	%	95	95	75
Mass flow rate	G	kg • s^{-1}	3.60	3.17	2.01
Turbine volume flow rate	$\dot{V}_{Tur}$	m^3 • s^{-1}	4.71	4.25	2.86
Compressor volume flow rate	$\dot{V}_{Com}$	m^3 • s^{-1}	1.71	1.61	1.31

Table 4
Optimal design parameters for the 40 g • mol^{-1} He-Xe Brayton cycle.

Parameter	Symbol	Unit	A	B	C
Cycle efficiency	η_e	%	36.46	36.16	26.98
Reactor inlet temperature	T_3	K	847.84	823.26	677.56
Pressure ratio	γ_{Com}	–	1.97	2.17	3.5
Recuperator effectiveness	α	%	95	95	75.2
Mass flow rate	G	kg • s^{-1}	34.17	30.72	19.44
Turbine volume flow rate	$\dot{V}_{Tur}$	m^3 • s^{-1}	4.62	4.24	2.86
Compressor volume flow rate	$\dot{V}_{Com}$	m^3 • s^{-1}	1.69	1.61	1.31

Table 5
Optimal design parameters for the 80 g • mol^{-1} He-Xe Brayton cycle.

Parameter	Symbol	Unit	A	B	C
Cycle efficiency	η_e	%	32.73	32.63	24.61
Reactor inlet temperature	T_3	K	838.58	824.76	677.50
Pressure ratio	γ_{Com}	–	2.03	2.14	3.5
Recuperator effectiveness	α	%	95	95	75.2
Mass flow rate	G	kg • s^{-1}	65.42	61.75	38.71
Turbine volume flow rate	$\dot{V}_{Tur}$	m^3 • s^{-1}	4.46	4.25	2.85
Compressor volume flow rate	$\dot{V}_{Com}$	m^3 • s^{-1}	1.64	1.60	1.31

Table 6
Optimal design parameters for the xenon Brayton cycle.

Parameter	Symbol	Unit	A	B	C
Cycle efficiency	η_e	%	24.31	24.31	19.24
Reactor inlet temperature	T_3	K	810.76	810.76	678.24
Pressure ratio	γ_{Com}	–	2.24	2.24	3.5
Recuperator effectiveness	α	%	95	95	75.4
Mass flow rate	G	kg • s^{-1}	95.15	95.15	63.03
Turbine volume flow rate	$\dot{V}_{Tur}$	m^3 • s^{-1}	4.01	4.01	2.82
Compressor volume flow rate	$\dot{V}_{Com}$	m^3 • s^{-1}	1.53	1.53	1.30

$$\boldsymbol{\eta_i} = [\eta_{Tur}, \eta_{Com}, \eta_{Mec}, \eta_{Gen}]^T \tag{30}$$

From Eq. (27), it can be observed that an increase in pressure ratio results in elevated temperatures at both the compressor outlet and the reactor inlet. At the same time, an increase in pressure ratio leads to a greater turbine expansion ratio, causing a decrease in turbine outlet and reactor inlet temperatures. Therefore, under the influence of these two opposing factors, the variation of reactor inlet temperature with pressure ratio is shown in Fig. 21. It is evident that the decrease in turbine outlet temperature with an increase in pressure ratio predominates, leading to a reduction in reactor inlet temperature. With a temperature ratio of 3.54, a recuperation degree of 90 %, and a total pressure loss rate of 8 %, when the pressure ratio increases from 1.5 to 1.97, the reactor inlet temperature decreases from 893.2 K to the upper limit of 823 K. Thus, to meet the reactor inlet temperature limit, the pressure ratio must be greater than 1.97.

After being preheated by the recuperator, the He-Xe enters the reactor to absorb heat; therefore, the recuperator effectiveness is a key factor affecting the reactor inlet temperature. The relationship between reactor inlet temperature and pressure ratio at different recuperator effectiveness is shown in Fig. 22. It can be observed that turbine outlet temperature is independent of the recuperator effectiveness. However, as the recuperator effectiveness increases, more preheating of the He-Xe from the turbine exhaust occurs, leading to a higher outlet temperature of the recuperator, which is the reactor inlet temperature, potentially exceeding the temperature limit. It can be seen that selecting a higher recuperator effectiveness and a lower pressure ratio can optimize cycle efficiency, but may lead to the reactor pressure vessel temperature exceeding the limit. Therefore, the thermodynamic performance of the GCMR Brayton cycle is constrained by engineering conditions such as the reactor inlet temperature.

4.3. Optimization of cycle thermodynamic performance

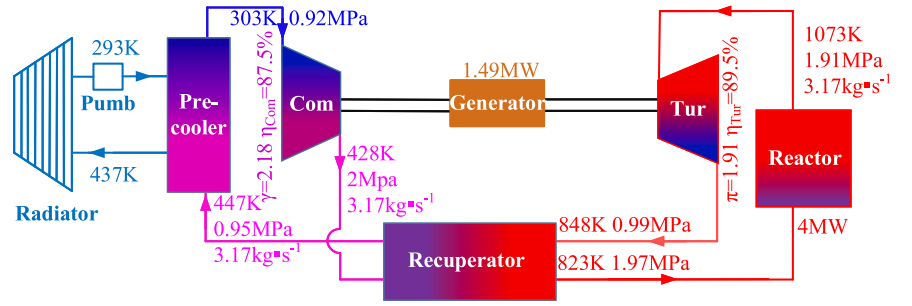
Based on the analysis in Sections 4.1 and 4.2, cycle efficiency and reactor inlet temperature are key performance indicators and constraints for the cycle. Certain parameter variations can lead to conflicting trends in cycle efficiency and reactor inlet temperature. For example, selecting a higher recuperator effectiveness and a lower pressure ratio can achieve higher cycle efficiency, but the increase in reactor inlet temperature can adversely affect the design of the reactor pressure vessel. Therefore, this section utilizes the Non-Dominated Sorting Genetic Algorithm (NSGA-II) for the optimization of cycle parameters [40]. Through multi-objective optimization of cycle efficiency and reactor inlet temperature, the Pareto front and optimal cycle design solutions are obtained.

In terms of the working fluid, there has been limited research on the influence mechanism of He-Xe compositions on Brayton cycle performance, and the selection of He-Xe compositions remains unclear. For example, the Brayton Rotating Unit (BRU) and Mini-BRU in the 1960s used a 83.8 g • mol^{-1} He-Xe as the working fluid [41–43]. In the 1980s, the French space nuclear power project ERATO proposed three systems using He-Xe with molar masses of 50 g • mol^{-1}, 55 g • mol^{-1}, and 60 g • mol^{-1} as the working fluid [44]. In the early 21st century, the JIMO and Prometheus projects used He-Xe with molar masses of 40 g • mol^{-1} and 31.5 g • mol^{-1} as the working fluid [45]. Based on the existing studies, this paper selects four working fluids: helium (4 g • mol^{-1}), 40 g • mol^{-1} He-Xe, 80 g • mol^{-1} He-Xe, and xenon (131.3 g • mol^{-1}) to optimize the system design.

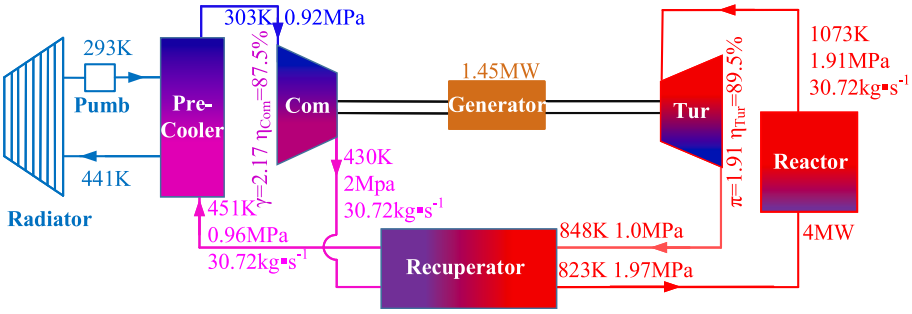
In terms of cycle parameters, considering the environmental temperature constraints and material thermal stress limitations, the temperatures at the compressor inlet and turbine inlet are 303 K and 1073 K, resulting in a temperature ratio of 3.54. The mechanical efficiency of the turbomachine has a minimal effect on the reactor inlet temperature and exhibits a monotonically positive impact on the cycle efficiency according to the analysis in Section 4.1. Therefore, the turbine efficiency and compressor efficiency are set to the current allowable design upper limits, at 89.5 %, and 87.5 %, respectively. Based on the analysis in Sections 4.1 and 4.2, the pressure ratio and recuperator effectiveness are key factors affecting the reactor inlet temperature and cycle efficiency, necessitating a multi-objective optimization study. The optimized objective and values of the parameters can be expressed as:

$$\eta_e = \max \eta_e(\tau, \alpha, \gamma_{Com}, \eta_{Com}, \eta_{Tur}, x_{He}) \tag{31}$$

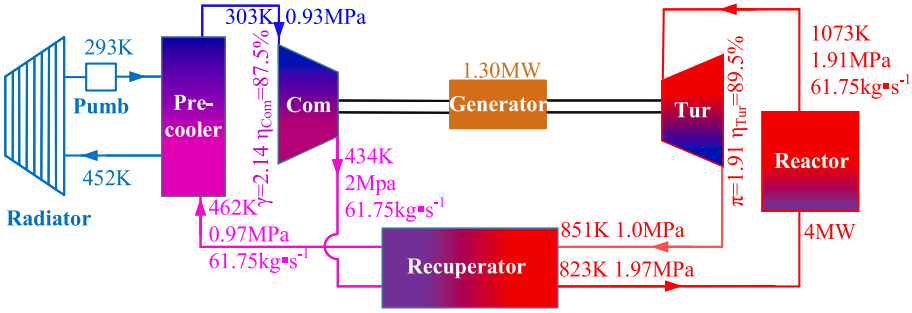
$$T_3 = \min T_3(\tau, \alpha, \gamma_{Com}, \eta_{Com}, \eta_{Tur}, x_{He}) \tag{32}$$



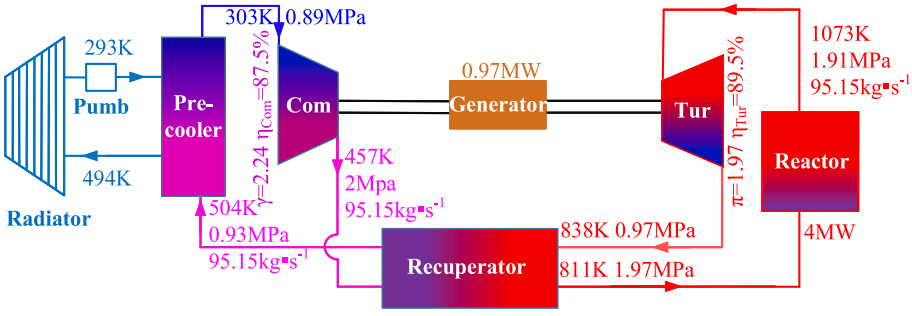
(a) Helium working fluid



(b) 40 $g \cdot mol^{-1}$ He-Xe working fluid



(c) 80 $g \cdot mol^{-1}$ He-Xe working fluid



(d) Xenon working fluid

Fig. 25. Cycle parameter design results.

$$\begin{cases} \tau = 3.54 \\ \eta_{\mathrm{Com}} = 87.5\% \\ \eta_{\mathrm{Tur}} = 89.5\% \\ 1.5 < \gamma_{\mathrm{Com}} < 3.5 \\ 75\% < \alpha < 95\% \end{cases} \tag{33}$$

The optimal boundaries for cycles using helium, 40 g • mol^{-1} He-Xe, 80 g • mol^{-1} He-Xe, and xenon working fluid are determined through optimization, as shown in Fig. 23. It can be observed that as cycle efficiency increases, the reactor inlet temperature also rises. The optimal ranges of cycle efficiency and reactor inlet temperature of these four working fluid compositions are [27.80 %, 37.70 %] and [677.86 K, 851.92 K], [26.98 %, 36.46 %] and [677.56 K, 847.84 K], [24.61 %, 32.73 %] and [677.50 K, 838.58 K], [19.24 %, 24.31 %] and [678.24 K, 810.76 K], respectively. The cycle efficiencies for the maximum efficiency scenario are 37.70 %, 36.46 %, 32.73 % and 24.31 %, with corresponding reactor inlet temperatures of 851.92 K, 847.84 K, 838.58 K and 810.76 K, all of which exceed the reactor inlet temperature limit, as indicated by point A in Fig. 23. The minimum reactor inlet temperature scenario results in cycle efficiencies of 27.80 %, 26.98 %, 24.61 % and 19.24 %, with corresponding reactor inlet temperatures of 677.86 K, 677.56 K, 677.50 K and 678.24 K, which are below the reactor inlet temperature limit, as shown by point C in Fig. 23. To achieve a higher cycle efficiency while adhering to the reactor inlet temperature limit, the optimal design point corresponding to the temperature limit of 823 K is selected as the optimal scheme. The cycle efficiencies for the helium, 40 g • mol^{-1} He-Xe, 80 g • mol^{-1} He-Xe working fluids under optimal schemes are 37.32 %, 36.16 % and 32.63 %, as indicated by point B in Fig. 23. The reactor inlet temperature of maximum efficiency scheme using xenon working fluid is 810.76 K below 823 K, therefore, the maximum efficiency scheme is the optimal scheme.

The optimal range for cycle parameters within the Pareto boundary is shown in Fig. 24. The optimal range for the pressure ratio is observed to be between 1.94 and 3.5, and the optimal range for the recuperator effectiveness is between 75 % and 95 %. Within these specified ranges, selecting a lower pressure ratio or a higher recuperator effectiveness can reach higher cycle efficiency; however, the reactor inlet temperature may exceed safety limits, as shown in Fig. 24(a) and 24(b).

Taking into account the power demands for the operation of a mobile GCMR, the reactor power is established at 4 MW. The key parameter values for helium, 40 g • mol^{-1} He-Xe, 80 g • mol^{-1} He-Xe, and xenon working fluids under conditions of maximum cycle efficiency, minimum reactor inlet temperature, and optimal design are listed in Tables 3, 4, 5 and 6. The power generations of the optimal cycle scheme for helium, 40 g • mol^{-1} He-Xe, 80 g • mol^{-1} He-Xe, and xenon working fluids are 1.49 MW, 1.46 MW, and 1.45 MW, respectively. The thermodynamic parameters of the components for these optimal schemes are shown in Fig. 25.

5. Conclusions

Thermodynamic performance and reactor pressure vessel temperature are critical performance indicators and constraints for the GCMR Brayton cycle. In this study, a mathematical model of the cycle was established. The influencing mechanisms of key parameters on power conversion performance were investigated, and considering the temperature limitations of the pressure vessel, the optimal design scheme for the cycle was obtained by maximizing cycle efficiency. The main conclusions derived from this study include:

(1) A thermodynamic model of a gas-cooled micro-reactor Brayton cycle is developed based on the variable properties of helium-xenon working fluid, and the primary factors influencing cycle performance include system parameters and working fluid properties.
(2) The helium- xenon working fluid composition is a key factor influencing the system's thermodynamic performance. For example, an increase in the helium fraction within the helium-xenon working fluid leads to changes in physical properties such as the heat capacity ratio and specific heat capacity, resulting in an increase in turbine output work and cycle specific power generation, as well as a monotonic increase in cycle efficiency.
(3) The thermodynamic performance of the cycle is constrained by the reactor pressure vessel temperature. For instance, selecting a lower pressure ratio and higher recuperator effectiveness can enhance cycle efficiency, but may lead to the reactor pressure vessel temperature exceeding the limit 823 K.
(4) The optimal thermal design schemes under the reactor pressure vessel temperature constraint were obtained through dual-objective optimization. For helium, 40 g • mol^{-1} helium-xenon, 80 g • mol^{-1} helium-xenon, and xenon working fluids, the cycle efficiencies of the optimal schemes under the reactor pressure vessel temperature constraint are 37.32 %, 36.16 %, 32.63 %, and 24.31 %, respectively.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. . Compressor outlet temperature and specific work

According to Eq. (1) through (4) in Section 3.1, the compressor outlet temperature and specific work can be expressed as follows:

$$T_2 = T_1\left[1 + \frac{C_{p,1-2s}}{C_{p,1-2}}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}} - 1\right)\right] \tag{A1}$$

$$w_{\mathrm{Com}} = T_1 C_{p,1-2s}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}} - 1\right) \tag{A2}$$

To standardize the form of the turbine and compressor equations and facilitate the derivation of formulas, the cycle temperature ratio τ is defined as the ratio of the highest cycle temperature (reactor outlet temperature) T_4 to the lowest cycle temperature (compressor inlet temperature) T_1: $\tau = T_4/T_1$. Substituting this into Eqs. (A1) and (A2), the compressor outlet temperature and specific work can be expressed as:

$$T_2 = T_4\tau^{-1}\left[1+\frac{C_{p,1-2s}}{C_{p,1-2}}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right)\right] \tag{A3}$$

$$w_{\mathrm{Com}} = T_4\tau^{-1}C_{p,1-2s}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right) \tag{A4}$$

Appendix B. . Specific power and inlet temperature of the reactor

The He-Xe working fluid from the compressor, enters the heat absorption side of the recuperator to absorb the residual heat from the turbine exhaust. According to the heat balance of the recuperator, the outlet temperature of the heat absorption side of the recuperator is:

$$T_3 = \alpha\frac{C_{p,2-5}}{C_{p,2-3}}(T_5-T_2)+T_2 \tag{B1}$$

According to the compression process of the compressor and expansion process of the turbine, the outlet temperatures of the compressor and turbine can be expressed as:

$$T_2 = T_4\tau^{-1}\left[1+\frac{C_{p,1-2s}}{C_{p,1-2}}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right)\right] \tag{B2}$$

$$T_5 = T_4\left[1-\frac{C_{p,4-5s}}{C_{p,4-5}}\eta_{\mathrm{Tur}}(1-\gamma_{\mathrm{Tur}}^{-\varphi_{4-5s}})\right] \tag{B3}$$

Substituting Eqs. (B2) and (B3) into Eq. (B1), the reactor inlet temperature can be expressed as:

$$T_3 = \alpha\frac{C_{p,2-5}}{C_{p,2-3}}T_4\left[1-\frac{C_{p,4-5s}}{C_{p,4-5}}\eta_{\mathrm{Tur}}(1-\gamma_{\mathrm{Tur}}^{-\varphi_{4-5s}})\right]+\left(1-\alpha\frac{C_{p,2-5}}{C_{p,2-3}}\right)T_4\tau^{-1}\left[1+\frac{C_{p,1-2s}}{C_{p,1-2}}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right)\right] \tag{B4}$$

The He-Xe from the heat absorption side outlet of the recuperator enters the reactor to absorb heat, and the specific power q_{Rea} of the reactor can be expressed as the difference between the specific enthalpies at the reactor outlet and inlet:

$$q_{\mathrm{Rea}} = C_{\mathrm{p},3-4}(T_4-T_3) \tag{B5}$$

Substituting inlet temperature of the reactor shown in Eq. (B4) into Eq. (B5), the reactor specific power can be expressed as:

$$q_{\mathrm{Rea}} = C_{\mathrm{p},3-4}T_4\left\{1-\alpha\frac{C_{p,2-5}}{C_{p,2-3}}\left[1-\frac{C_{p,4-5s}}{C_{p,4-5}}\eta_{\mathrm{Tur}}(1-\gamma_{\mathrm{Tur}}^{-\varphi_{4-5s}})\right]-\left(1-\alpha\frac{C_{p,2-5}}{C_{p,2-3}}\right)\tau^{-1}\left[1+\frac{C_{p,1-2s}}{C_{p,1-2}}\eta_{\mathrm{Com}}^{-1}\left(\gamma_{\mathrm{Com}}^{\varphi_{1-2s}}-1\right)\right]\right\} \tag{B6}$$

Appendix C. . The physical properties of the helium-xenon working fluid

The formula for calculating the average molar mass of the helium-xenon is:

$$M = x_{\mathrm{He}}M_{\mathrm{He}}+(1-x_{\mathrm{He}})M_{\mathrm{Xe}} \tag{C1}$$

where M_{He} and M_{Xe} are the molar masses of the helium and xenon.

The calculation formula for the thermophysical properties of the helium-xenon is obtained using the truncated Virial equation. First, the density is as follows:

$$\rho = \frac{p}{RT\left[1+B\frac{p}{RT}+C\left(\frac{p}{RT}\right)^2\right]} \tag{C2}$$

where R is the gas constant, and T and p represent the temperature and pressure of the gas, respectively.

The specific heat capacity of the helium-xenon is expressed as:

$$C_P = C_P^0+\rho R\left[\left(B-T\frac{dB}{dT}-T^2\frac{d^2B}{dT^2}\right)+\rho\left(C-\frac{T^2}{2}\frac{d^2C}{dT^2}\right)\right]+RT\left[\left(B-T\frac{dB}{dT}\right)+\rho\left(2C-T\frac{dC}{dT}\right)\right]\times\left[\frac{\partial\rho}{\partial T}\right]_p \tag{C3}$$

$$C_V = C_V^0-\rho R_gT\left[\left(2\frac{dB}{dT}+T\frac{d^2B}{dT^2}\right)+\rho\left(\frac{dC}{dT}-\frac{T}{2}\frac{d^2C}{dT^2}\right)\right] \tag{C4}$$

where $C_P^0=\frac{5R}{2}$ and $C_V^0=\frac{3R}{2}$ represent the specific heat capacity for a monoatomic ideal gas; B and C are the second and third virial coefficients for the helium-xenon mixture, with the following calculation:

$$B = x_{\mathrm{He}}^2B_{11}+2x_{\mathrm{He}}(1-x_{\mathrm{He}})B_{12}+(1-x_{\mathrm{He}})^2B_{22} \tag{C5}$$

$$C = x_{\mathrm{He}}^3C_1+3x_{\mathrm{He}}^2(1-x_{\mathrm{He}})C_{112}+3x_{\mathrm{He}}(1-x_{\mathrm{He}})^2C_{112}+(1-x_{\mathrm{He}})^3C_2 \tag{C6}$$

where B_{11} and B_{22} are the second virial coefficients for the pure gases; B_{12} is the second virial interaction coefficient for the helium-xenon; C_1 and C_2 are the third virial coefficients for the two gases; $C_{112} = \left(C_1^2 C_2\right)^{1/3}$ and $C_{122} = \left(C_2^2 C_1\right)^{1/3}$ are the third virial interaction coefficients for the helium-xenon.

The second virial coefficient for helium and xenon are calculated as follows:

$$B_{11} = \left(8.4 - 0.0018T + \frac{115}{\sqrt{T}} - \frac{835}{T}\right) \times 10^{-6} \tag{C7}$$

$$B_{22} = V_2^* \left[-102.6 + \left(102.732 - \frac{\theta_2}{1000} - \frac{0.44}{\theta_2^{1.22}}\right) tanh\left(4.5\sqrt{\theta_2}\right) \right] \tag{C8}$$

where V_2^* is the critical molar volume of xenon; $\theta_2 = \frac{T}{T_{cr,2}}$ is the relative temperature, $T_{cr,2}$ is the critical temperature of the xenon.

The second virial interaction coefficient B_{12} of the helium-xenon is expressed as:

$$B_{12} = V_{12}^* \left[\left(102.732 - \frac{\theta_{12}}{1000} - \frac{0.44}{\theta_{12}^{1.22}}\right) tanh\left(4.5\sqrt{\theta_{12}}\right) - 102.6 \right] \tag{C9}$$

where V_{12}^* is the critical molar volume of helium-xenon; $\theta_{12} = \frac{T}{T_{cr,12}}$ is the relative temperature, $T_{cr,12}$ is the critical temperature of the helium-xenon.

The third virial coefficient of helium or xenon is as follows:

$$C_i = \left(V_i^*\right)^2 \left[0.0757 + \left(-0.0862 - 3.6 \times 10^{-5}\theta_i + 0.0237/\theta_i^{0.059}\right) tanh(0.84\theta_i)\right] \tag{10}$$

where V_i^* is the critical molar volume of helium or xenon; θ_i is the the relative temperature of helium or xenon.

Data availability

No data was used for the research described in the article.

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